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VRU Safety at Road Intersections: An Ontology-Driven Digital Twin

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ABSTRACT Although the role of optimization in transportation systems is widely recognized as a key to future sustainable development, alarming statistics related to accidents and deaths continue to be overlooked. Among the various components in road networks, road intersections pose the greatest safety challenges for smart and human cities. Digital twins (DT) have emerged as a promising alternative to assist ITS's (Intelligent Transportation Systems) decision-makers in recent years. However, a critical component is still lacking in developing DTs for ITS: a robust information modeling framework that effectively integrates the physical world with cyberspace in these intricate and dynamic environments. Furthermore, the current emphasis on the efficiency of ITS solutions often overshadows the pressing safety concerns for vulnerable road users (VRUs). This paper tackles the challenge of modeling transportation scenarios to enhance VRU safety, focusing on a crucial element: a *road intersection*, in particular, the semantic issues of such elements when addressing VRU safety. For the first time, we propose an ontology-based information modeling method for road intersections using recent advancements from various market-driven, regulatory, and standardization bodies. It specifies the information classification, the content to be modeled, and the modeling methodology. Nevertheless, inexpensive, swift, and friendly prototyping is still a bottleneck for innovation in this area. Practitioners currently need more means and methods to realize their ontology-based DTs for ITS. Thus, another contribution is to provide an information model instantiation, visualization, and simulation, as well as fully-based open-source tools, as a case study.

INDEX TERMS digital twins, information modeling, ontology, road intersection, vulnerable road users, FIWARE, CARLA Simulator

I. INTRODUCTION

THE role of transportation systems is critical for modern society's sustainable development. The United Nation's 2030 Agenda presents 17 sustainable development goals (SDGs). Sustainable transport is related to i) the care of life and the well-being of people through the road safety approach (SDG 3.6), ii) the reduction of polluting emissions (SDG 3.9), iii) the creation and use of sustainable infrastructures (SDG 9.1), and iv) the conception of sustainable transport systems

that allow access to transport in a safe, affordable way for all, improving road safety (SDG 11.2). These goals explicitly show the concern for current transport problems and road safety [1].

Vulnerable road users (VRU) are central in intelligent transportation systems (ITS) [3]. VRUs are formally defined as "non-motorized road users, such as pedestrians and cyclists as well as motorcyclists and persons with disabilities or reduced mobility and orientation" [4]. The likelihood of

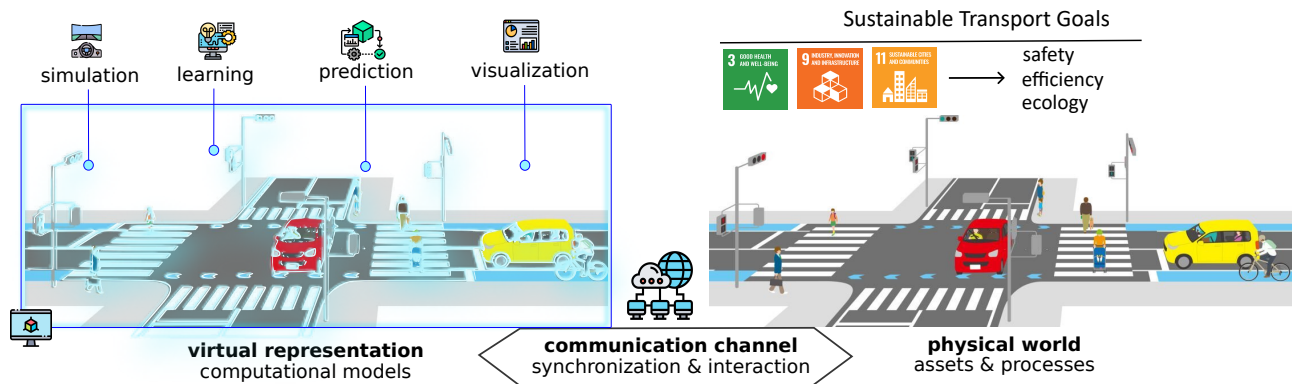


FIGURE 1. Digital Twins Paradigm. [2]

a pedestrian suffering serious consequences from a collision event is almost 300 times more than drivers' [5]. However, VRU safety is actually dependent on multiple players, not only on drivers. Outdated roadside infrastructure, technological limitations of vehicles, and even inconsequential behaviors from VRUs themselves are also to blame for 20 to 50 million accidents with non-fatal injuries, and 1.19 million fatal ones, at road traffic crashes every year, according to statistics from the World Health Organization (WHO) [6]. This situation drives efforts toward VRU-centered ITS designs, and further research is fundamental at road intersections, where *these participants (including VRUs, drivers, and vehicular systems)* will interact closely. Note that they will act and react to each other's strategies for dynamically sharing a physical space, where occlusions and multiple blind spots are present like no other road segment. Among other measures involving VRU-awareness algorithms, collision avoidance in the conflict zones, at roadside equipment, or on virtual assistants embarked into vehicles, is urgently needed to improve VRU safety.

Note that road intersection scenarios prove to be very complex in terms of semantic representation of many agents and interaction scenarios. Agility for acting and preventing unwanted states involving VRUs in transport systems, such as the ones recently defined by [7], is turning into an elusive goal in spite (or perhaps because) of recent technological advances. The proliferation of data and diverse data formats from roadside equipment, transferred using various protocols and processed either at the edge or in the cloud with advanced AI capabilities, along with the emergence of connected and automated vehicles, introduces new challenges for network routing.

Alternatively, digital twins (DT) may not only help ITS to deal with such semantic complexity but also unlock its wealth. The current abundant information in ITS and the latent potential for providing better public services need to be unleashed. In a nutshell, as illustrated in Fig. 1, the key architectural concept here is that a given physical asset can be

properly mapped into a digitized counterpart entity in DTs. They use a communication channel (and proper methods) to enable their physical and virtual parts to interact and co-evolve with each other [8]. Various sensors and scanning technologies, entities, actuators, behaviors, and knowledge about relationships in the physical world are used to create high-fidelity virtual models. Several well-known problems in transportation systems, such as safety, and efficiency, can be addressed in favor of the sustainability of such systems. Virtual models rely on real-world data to formulate their real-time parameters, conditions, and dynamics, improving the likelihood of reflecting the corresponding physical entities. With such models, it would be possible to simulate and analyze *what-now*, *what-next*, and *what-if* scenarios to obtain quick and trustful answers, allowing act accordingly on physical transport scenarios. In other words, DT may foresee problems and be proactive across different dimensions of ITS, including when dealing with safety issues for VRUs. In this context, digital twins can play the role of technology orchestrators in enabling VRU risk mitigation solutions.

On the other hand, the downside of DTs lies in the fact that modeling a transport vertical application is still heavily domain-dependent, i.e., it requires deep knowledge of the application intricacies for properly implementing a DT [9]. The development of DT-based transport applications is still in the very early stages, and this is partly due to the absence of effective methodologies for representing their knowledge domains in proper semantic models. This paper argues that developing ontology-driven modeling is now crucial to pushing forward DT solutions in next-generation ITS applications.

Ontology engineering is here understood as the representation, naming, and adequate definition of the categories, properties, and relations between the concepts, data, and entities of a given domain of interest [10]. In DT, the proper modeling of a system's physical (or virtual) entities will be useful for understanding its parts and duly representing the system's complex interactions. Thus, the use of domain ontologies also facilitates interoperability and exchange of information

between different DT components [11].

This paper proposes, for the first time, a pragmatic yet comprehensive approach involving ontology to model the intricacies of physical systems and implement a smart data model for road intersection scenarios focusing on VRU issues in DTs. Note that this ontology-based approach may serve as a template for DT capabilities to fulfill the "data services module" by the Digital Twins Consortium [12]. Within such modules, DTs for road intersections, as well as other segments, can manage knowledge graphs and ontologies, allowing the DT-based solution to interpret heterogeneous data directly from the transport domain ontology as well. Thus, structured information can be exchanged and shared between different stakeholders.

Furthermore, efforts to unlock the instantiation of DTs in ITS, besides addressing semantic issues, are also needed. Our proposal connects: i) Recent advances in collision cases definitions according to the European New Car Assessment Program [7]; ii) Information models such as next-generation service interfaces-linked data (NGSI-LD) [13] for publishing, querying, and subscribing to context information; and finally, iii) the use of a market-driving open-source platform, i.e., FIWARE, to demonstrate implementation of such data models so that compatibility with IoT and other context information management and big data services in the cloud can be taken into consideration [14].

The remainder of this work is organized as follows. Section II discusses the semantic issues in digital twins for ITS, highlighting the main gaps in creating DT information models in ITS. Section III introduces The Road Intersection Ontology, a domain ontology to perform reasoning that allows DT-based intelligent road intersections (DT-RI) to implement applications to improve the VRU's safety. Section IV outlines the use of the proposed ontology to develop the smart data model and subsequently instantiate road intersection digital twins within the FIWARE framework. Finally, Section V concludes the paper and proposes future works.

II. THE SEMANTIC ISSUES IN DIGITAL TWINS FOR ITS

Digital Twins for Intelligent Transportation Systems, and specifically for intelligent intersection solutions, can be broadly defined as the digital representation of the diverse components, including i) the intelligent infrastructure (roads, traffic lights, and roadside elements in general), ii) the traffic participants (vehicles, non-vehicles, and vulnerable road users), iii) the traffic behavior (departure, destination, path, etc.), and iv) the surrounding environment (urban functional area, weather conditions, etc.). A successful representation and reliable virtual models significantly improve common problems of transportation systems, including safety in road intersections, even with the early discovery of such issues. Thus, short-, medium-, and long-term strategies can be designed to support decision-making at intersections, trans-

forming them into safer environments for VRU.

It is widely believed that gathering more comprehensive information from roads, vehicles, and drivers can significantly enhance awareness of hazardous traffic and road conditions. Depending on weather conditions and traffic congestion, they may become adverse to VRU and drivers. It's expected that diverse verticals will tap into this wealth of information for safety and efficiency in ITS. As depicted in Fig. 2, **real-time and detailed road and traffic information may allow road authorities to exercise more precise traffic control via new adaptive and dynamic smart traffic light systems to improve safety and traffic efficiency at road intersections [15], [16]. By utilizing diverse IoT devices, including wearable devices for VRU and drivers, combined with digital maps and road-building information models, systems can be developed to monitor road traffic systems, assess road health, and track infrastructure assets effectively. By utilizing diverse IoT devices, including (wearable) devices for VRUs and drivers, combined with digital maps and road infrastructure information models, it is possible to build accurate digital twins of road traffic systems and infrastructure assets. These digital representations enable real-time monitoring and effective analysis, such as event prediction, which are crucial for the safety [17] and the efficiency [18] of intelligent transportation systems.**

However, there are yet considerable gaps in the current structured understanding of data and flow representation and reasoning layers in the transportation verticals [19]. No best practices have been established for organizing such a system as a proper DT for road intersections, which is possibly the most critical element of ITS. We argue that only well-designed ontologies can effectively address such complexity. For instance, dividing the system's static information, such as the geometric infrastructure of the road network and the various situations, from dynamic elements of the system provides a logical basis for reasoning about the system's behavior, which can be represented more accurately in the data model used by the digital twin in its prediction, analysis, and visualization tasks.

Regular state graphs, which are often used in software engineering to represent the behavior of complex systems, like the Unified Modeling Language (UML), perhaps disadvantageous as the number of situations to be considered in ITS grows, basically due to its closed world assumption (CWA) [5]. In other words, UML assumes a complete representation of all the knowledge in the domain, whereas a semantic model based on ontology tolerates partial knowledge about a domain; so, in case a statement cannot be inferred as true or false about an object, it is assumed to be unknown. Thus, ontologies not only allow unknown system situations but also a kind of semantic modeling that learns from the complexity by accumulating knowledge observed in new scenarios. This is why the large community dealing with artificial intelli-

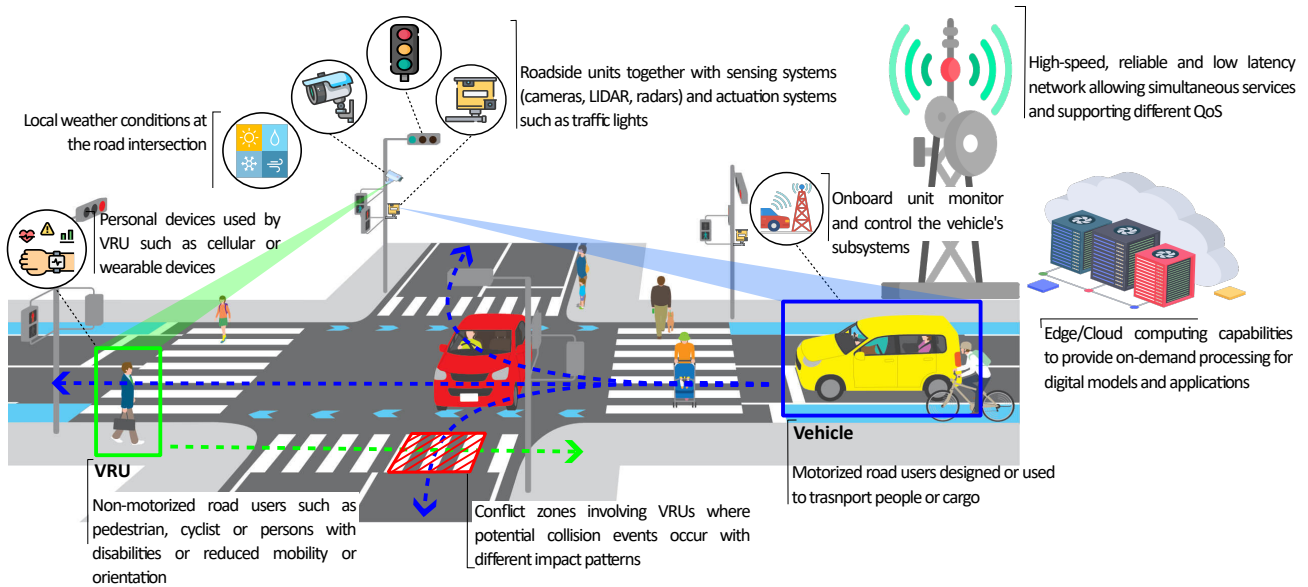


FIGURE 2. Elements for information gathering, transporting, and processing for ITS unified management and control over roads, vehicles, and human interactions

gence (AI) subscribes to ontologies as a preferred model of knowledge representation and reasoning [20]. Recently, there have been initiatives designed to represent and integrate taxonomies, such as Common Core Ontologies (CCO), toward generic classes and relations across different domains of interest for an integrated, consistent, and semantically interoperable monolithic piece [21]. For further discussions on DT interoperability and the central role of ontologies in semantically uniting DT and AI for building cognitive DTs, readers are referred to [19].

A. RELATED WORKS

In the context of transportation systems, it is worth highlighting a review of early efforts focused on systematic and reliable methods for capturing semantic information through ontology [22]. The proposal of sustainability metrics, with dynamic features, for the coverage of open fleets can be found in pioneering ITS ontologies [23]. There is a recent review on ontology-based approaches for diverse issues related to smart city services [24]. In the context of identifying elements and the effort to group common characteristics into categories, a taxonomy is established for modeling and developing new context-aware ITS [25].

For the structured categorization of traffic environments, Scholtes *et al.* proposed a 6-Layer Model (6LM) [26] that defines the elements from the road network and roadside structures to the dynamic and digital elements of the traffic environment. The 6LM has been widely used since its publication in different proposals in the transport domain, largely because it is a formal basis for building ontologies using the Web Ontology Language. However, the model is limited to the elements of each layer and does not address the properties

and relationships between elements within and between layers. Specifically, to meet safety goals, in Westhofen *et al.* [27], a 6LM-based ontology for modeling and formalizing critical factors associated with autonomous vehicle environments is proposed. The authors created the Automotive Urban Traffic Ontology (A.U.T.O.) with a modular approach to enable reasoning about criticality in urban traffic scenarios, which was evaluated using a large-scale drone data set of traffic scenarios. Based on the A.U.T.O. ontology, Zhang *et al.* made further extensions and adjustments to obtain a more complete traffic scenario perspective, updating the ontology based on the fusion results of the small-scale knowledge graph [28].

Also focusing on safety in traffic scenarios, Reich *et al.* [29] proposed a metamodel for cross-domain safety analysis for autonomous vehicle systems. The model allows the creation of Operational Domain Design (ODD) in a structured manner, enabling cross-domain learning and representing a starting point for ontological modeling of the transport domain from a safety engineering perspective. Ruchiga *et al.* [30] proposed a generic ontology and framework for describing and generating critical scenarios. The proposed ontology combines the description of the scenario, environment, and the dynamic elements' condition and behavior for both humans and autonomous vehicles, all of which are needed to have a successful mixed traffic scenario model. Although this last work brings one of the most complete ontologies found in the literature for the analysis of safety-critical scenarios, the methods for its implementation in real data platforms are not described. Another ontology for collaborative navigation in traffic scenarios involving autonomous vehicles, drivers, and pedestrians was proposed by Syzydkbayev *et al.* [31], but yet

again without the corresponding implementation through data models.

Associating perceived information with contextual information to infer non-safety situations in traffic scenarios can be complex. In Armand *et al.* [32], an ontology-based context awareness framework was proposed to understand and interpret a context when perceived by vehicle sensors. In this solution, the ontology permits the establishment of different relationships between the road entities to infer their behavior in both medium and long terms. Besides, a long-term prediction framework, which considers the chain reactions among road users, was proposed by Fang *et al.* [33] using an ontology-based reasoning approach. The proposed ontology was defined with a knowledge base that can be applied to various traffic scenarios and reduce the computation time compared to traditional methods.

Although physical assets in a transport system, such as infrastructure, vehicles, or even pedestrians, are modeled based on ontologies, there was a gap in representing vulnerable road users in traffic scenarios. To address the lack of descriptive language for road users and drivers, Dodoiu *et al.* [34] proposed a methodology for incorporating human factors into scenario descriptions for testing autonomous driving systems. The methodology considers a wide range of characteristics relevant to human road users but does not address the analysis of their interactions. The challenge of modeling comprehensive VRU data and their interactions with other elements of the transportation system remains an open issue.

In summary, Table 1 brings three main features discussed in this work, compared with the set of related works described above to highlight our contributions. Here, we are interested in i) Whether VRU is considered in their ontology approach as an important element within the transport system, ii) To what extent a corresponding information model for the ontology is proposed, and finally, iii) whether an implementation for their information model is presented according to market-driven platforms.

Although VRU must be a central element of future ITS solutions, Table 1 highlights gaps in such ontology-based approaches as VRU's safety issues are only partially addressed. Moreover, most studies fail to propose information models for their ontologies. In this field of expertise, abstract speculations are usually preferred rather than spending energy figuring out how such abstractions could be brought closer to real-life implementations by using current enabling technologies. It is also remarkable that important efforts, such as market-driven test protocols regarding VRUs, are overlooked when implementing DT-ITS data models.

While ontology-driven modeling is a necessary step toward next-generation ITS applications, it is not sufficient on its own. Proper data models and their compatibility with de facto industry standards for providing meaningful and swift prototype implementations are also important to unlock the

TABLE 1. Summary of the evaluation of related works and our proposal.

Reference	VRU-centric ontology	Information model	Market-driven implementation
Scholtes <i>et al.</i> [26]	●	○	○
Westhofen <i>et al.</i> [27]	●	○	○
Zhang <i>et al.</i> [28]	●	○	○
Reich <i>et al.</i> [29]	○	●	●
Ruchiga <i>et al.</i> [30]	●	○	○
Syzdykbayev <i>et al.</i> [31]	●	○	○
Armand <i>et al.</i> [32]	●	○	○
Fang <i>et al.</i> [33]	○	○	○
Dodoiu <i>et al.</i> [34]	●	●	○
Our proposal	●	●	●

virtuous cycles of innovation. This work, for the first time, bridges across these three phases. A DT road intersection use case demonstrates such a prototyping process.

III. PROPOSAL FOR AN ONTOLOGY FOR VRU SAFETY IN A ROAD INTERSECTION DOMAIN

For the accurate and timely development of DT-based applications in intelligent road intersections (DT-RI), we have proposed a systematic framework composed of four layers due to extensive surveying work [35]. At the bottom of the framework, the physical layer for intelligent intersection solutions can be defined as the digital representation of the intersection components, including the smart infrastructure, the traffic participants, the traffic behavior, and the surrounding environment. These physical entities serve as data sources for the virtual models. The definition of physical entities within the DT standards varies markedly between domains as they are adapted to specific application scenarios. To accelerate the development and adoption of DT-based transportation applications, using layer or dimension ontologies of the DT-RI framework [36] is critical.

The Data Layer and Model Layer are where the semantic module of the DT-RI framework is implemented, as shown in Fig. 3. The aggregated data from the Physical Layer is stored in the cloud to perform semantic operations in the Data Layer. To achieve semantic interoperability, the semantic module must comprise ontologies to provide semantic annotations to the aggregated data and convert the aggregated data into semantic data to create the virtual models in the Model Layer for instantiating our DT-RI.

A. THE INFORMATION MODELING METHOD OF ROAD INTERSECTION DATA

Road intersections have heterogeneous data from multiple sensors dispersed throughout the scenario of interest. When discussing information modeling, the common practice is to create an ontology model written in a language that supports semantic richness, enabling a greater capacity for exchanging machine-readable content. Thus, the fundamental concepts of road intersection operations to improve the safety of VRUs

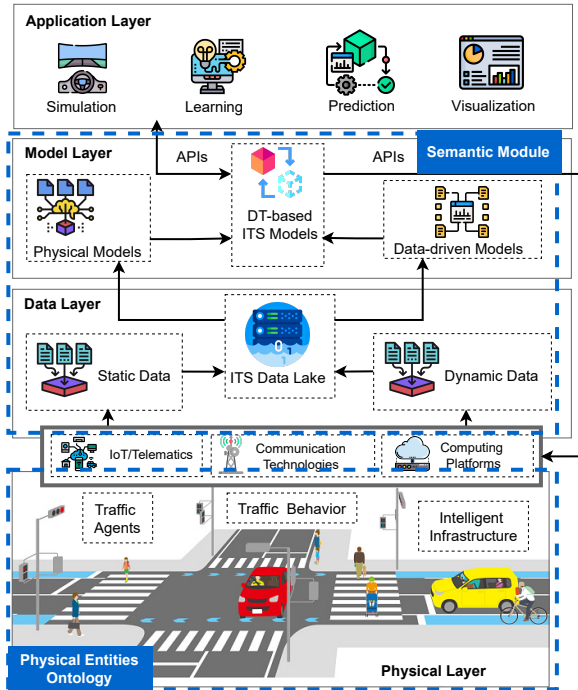


FIGURE 3. Semantic Module in the DT-ITS reference framework. Adapted from [35].

can be defined clearly and unambiguously through the formal description of entities and their relationships.

A DT-RI is a set of virtual information constructs that fully describe an actual physical road intersection. In the bottom layer of the DT-RI framework, the presence of the ontology is essential since it plays a critical role in enabling the effective modeling of the DT-based road intersection. At this layer, a well-defined domain ontology enables the accurate representation of physical entities for the DT-based road intersection.

Our Road Intersection Ontology proposal defines more precisely what the DT's virtual entity will be concerned with regarding the concepts in the domain, their relationships, properties, and possible constraints. We aim to identify the main concepts and relationships in the transportation system, specifically road intersections, to address VRU safety DT-based applications. The ontology model can be described as a set of concepts or classes, individual instances, and the relations between classes. After analyzing the existing ontologies in the intelligent transportation domain, we propose a domain ontology to perform reasoning that allows DT-RI to implement applications to improve the safety of VRUs.

Our proposal is based on the SAREF4AUTO core [37] ontology as the primary reference, a description logic ontology using Web Ontology Language (OWL) focused on the automotive domain. In addition, new elements have been incorporated for the road intersection scenario to meet many

dangerous situations for VRU elements.

The ontology is structured modularly using a semantics-driven approach [38]; this enables expressivity requirements to be easily handled and provides scalability for the evolution and maintenance of the ontology. As the base cell of the road network in urban and non-urban areas, every road intersection has the same structural information: dynamic and static. Once both the dynamic and static information modeling of the road intersection are detailed, combining these two parts can create a complete DT-based road intersection. The ontology was implemented and evaluated using the Protégé software¹. The evaluation demonstrates that the presented ontology can process human-like reasoning in road intersection contexts. For the visualization of the ontology, we have used the OWLGrEd software².

For the sake of clarity, our ontology proposal has been divided into two main modules, the static physical information of the road intersection and the dynamic physical information of the road intersection. These modules are formed by different terminology classes that are merged into a complete ontology. In Fig. 4 and Fig. 5, squares denote classes; straight blue arrows represent the subclass relationship between two classes (the source of the arrow is the subclass of the destination class of the arrow); and dashed red arrows represent OWL properties used to assert relationships between classes (or instances).

B. INTERSECTION STATIC INFORMATION MODELING

The road intersection static ontology is defined in Fig. 4 for mapping non-moving elements in Fig. 2 into their DT representation. At the center of the topology, the *Intersection* class was created to define the main objects of interest for VRU safety. This class is related to the *Topology* class through the *consistsOf* property. The road topology comprises the spatial objects at the intersection, including their geometric features and defining spatial relations among the objects. The property *consistsOf* describes the spatial entities that can compose the intersection, while the property *belongsTo* describes the administrative area where the road intersection is placed. Since DT-RI's will be part of macro smart city solutions, unique identification of such road intersections is necessary so that the authorities responsible for the smart intersection can detect and isolate faults and perform management tasks efficiently. The *TrafficCenter*, the *State*, *City*, *Neighborhood*, are all subclasses of the *AdministrativeArea* class. The *TrafficCenter* class serves as the central hub for monitoring key actions related to VRU safety. It provides services to road users, facilitates data-driven road traffic management, and provides essential information about mobility and the operational management of road intersections.

¹<https://protege.stanford.edu/>

²<http://owlgred.lumii.lv/>

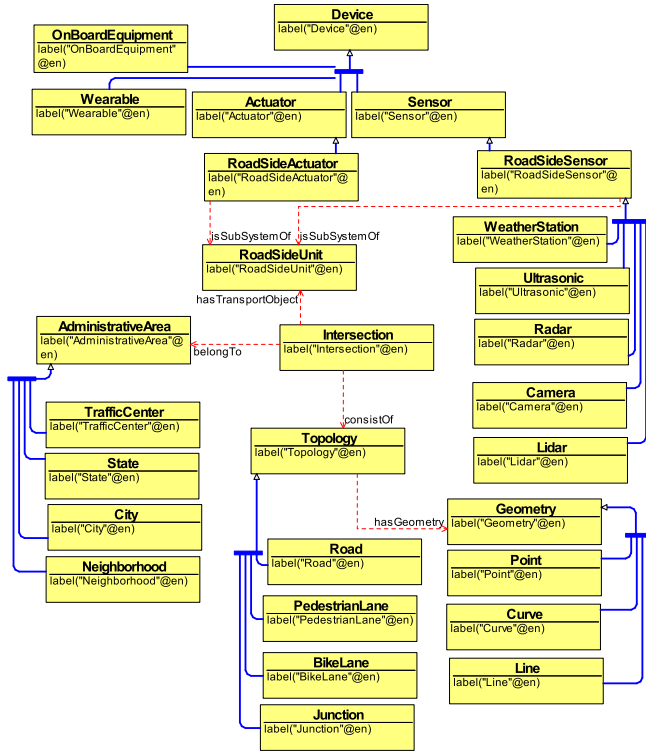


FIGURE 4. Static information modeling for road intersection digital twins.

The *Topology* class has been created to represent the static characteristics of the intersection. Some aspects of the GeoSPARQL [39] ontology have been reused to represent spatial objects, defining their spatial features and geometries. The road intersection has some geometry described by the *hasGeometry* property. The *Geometry* class represents any spatial entity, being or having a shape, position, or extent. Any of these entities at the road intersection has features related to a *Geometry*, based on *Point*, *Line*, and *Curve* classes in the road intersection space. A point is a geometric object representing a single location in a Cartesian coordinate space. A Line is a curve with linear interpolation between exactly two points, and a curve is a geometric object stored as a sequence of points. They all serve as a reference for describing the topological features of the road intersection, e.g., roads, lanes, and junctions.

As an important part of the DT-RI, simulations employ software that requires a detailed topological representation of the civil characteristics of the scenario to be modeled. Explicitly understanding the semantics of this information is essential for constructing more realistic worlds used in simulations for VRU safety applications. For this are defined the i) *Road* class, a broad pathway leading from one location to another, typically with a specially designed surface for vehicle use; ii) *BikeLine* class, a designated section of a road intended for use by cyclists; iii) *PedestrianLane* class, a walkway on one or both sides of a road designed for pedestrians; and iv) *Junction* the point where roads intersect or converge. All this

topological information allows us to understand the global context of the road intersection, allowing the participants of the transportation system to be located (including the VRUs) to execute simulations involving such participants to analyze and hopefully predict and resolve eventual safety conflicts among them.

The *Device* class, defined in the SAREF Ontology [40], has been reused to specify sensors and actuators specific to transportation systems. Then, the *RoadSideSensor* and *RoadSideActuator* classes are related through the *isSubSystemOf* property with the *RoadSideEquipment* class which is a main part of the road intersection. Roadside sensors, including cameras, radars, lidars, and ultrasonic devices, are deployed along intersections to monitor the behavior of traffic agents, as illustrated in Fig. 2. The actuators are devices deployed along the intersection that produce signals based on the output of the analyses carried out by specific DT-based applications. It is also considered the Electronic Control Unit (ECU), part of the vehicle's onboard equipment. For this, the *OnBoardEquipment* class is created, being a subclass of the *Device* class. The onboard equipment is any embedded system in automotive electronics that monitors and controls one or more of a vehicle's electrical systems or subsystems. It can use various and heterogeneous sensors and devices in the vehicle to provide measurements and control properties. Moreover, wearable devices primarily used by the VRU are also considered. Passengers or drivers of vehicles may use personal devices, but they are considered different devices from the vehicle onboard unit.

C. INTERSECTION DYNAMIC INFORMATION MODELING

An ontology model needs to address also the challenge of representing dynamic characteristics of the road intersection, illustrated in Fig. 2, into representative elements of a DT's virtual world. As expected, the complexity in Fig. 5 organically reflects the challenge faced by an ontology focused on dynamic scenarios related to VRU safety.

The *Intersection* class is related to the *TrafficAgent* and *Weather* classes through the *hasTrafficAgent* and *hasWeather* properties, respectively. These last two classes represent the dynamic elements of interest in the road intersection to develop VRU safety applications. The *TrafficAgent* class represents any entity present in the intersection, including any vehicle type and the VRUs. For any traffic safety application, the vehicle and VRU must be located in the intersection topology from the static information modeling, for which the *hasAbsolutePosition* and *hasRelativePosition* properties are created. The *hasRelativePosition* property is the relation between a traffic agent to represent their relative position in front of a VRU, and the *hasAbsolutePosition* property is the relation between a traffic agent to represent its absolute position in the road intersection scenario. The *TrafficAgent* class has the *hasaRoute* property, a relation to express the

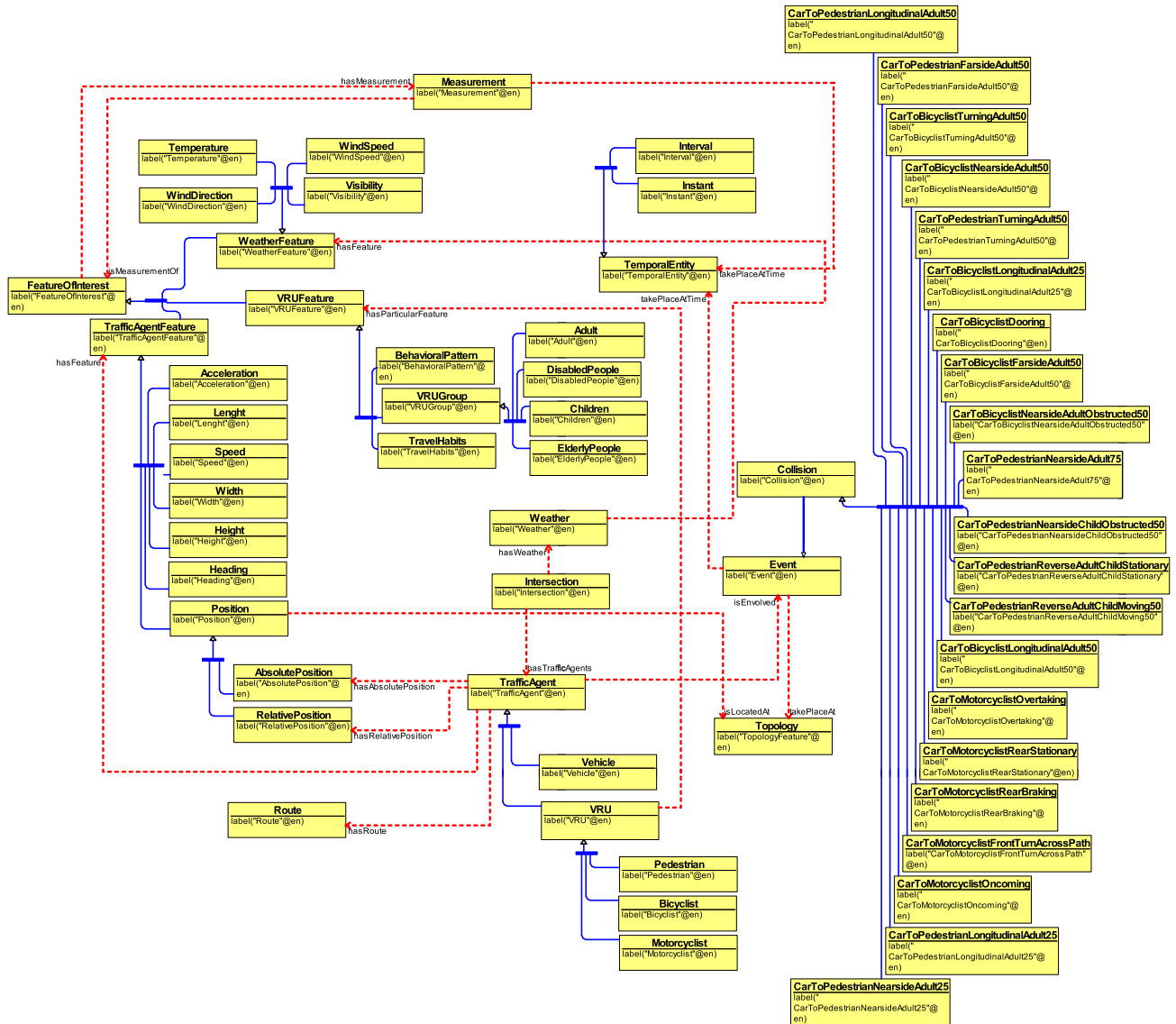


FIGURE 5. Dynamic information modeling for road intersection digital twins.

route of an entity, e.g., a vehicle or a VRU, whose route is defined by the *Route* class in the SAREF4AUTO ontology.

The *VRU* class describes road users, such as pedestrians, cyclists, motorcyclists, and other persons with disabilities or reduced mobility and problems with orientation. We have classified the VRUs into various profiles, with i) profile 1 being the *Pedestrian* class, which includes adults, children, elderly persons, blind persons guided by a dog, and riders off their bikes; ii) profile 2 the *Bicyclist* class that includes users of bicycles, wheelchairs, skaters, scooters, or horse riders; iii) profile 3 the *Motorcyclist* class that addresses users of two wheels vehicles which are equipped with engines that allow them to move on the road.

The DT-RI obtains the physical scenario data through sensors (described in the static information ontology in Fig. 4) to keep the digital models synchronized and the actuators to intervene in the physical road intersection represented by a DT.

The sensors provide the necessary measurements to obtain the features of interest for creating DT-RI instances used by VRU safety applications. The *FeatureOfInterest* class covers all the features of the road intersection divided into the subclasses *TrafficAgentFeature*, *VRUFeature* and *WeatherFeature*. The *TrafficAgentFeature* class considers the common mobility characteristics of traffic agents such as speed, acceleration, and heading, among others, about the traffic agents' geometry like length, width, and height. In addition, the *VRUFeature* class includes all the specific features of the VRUs, such as the behavioral pattern, the groups belonging to the VRU, and their travel habits.

Specifically, events that involve any participants of the transit are monitored through the *Event* class of the SAREF4AUTO ontology, using the *takePlaceAt* property to define the relationship between an event and the road topology facility in which it takes place and the *takePlaceAtTime*

TABLE 2. Summary of VRU collision cases (adapted form [7]).

Class	Definition
<i>CarToBicyclistDooringAdult</i>	A collision between the vehicle's door or an occupant exiting a vehicle and a bicyclist traveling alongside the vehicle.
<i>CarToPedestrianFarsideAdult50</i>	An adult pedestrian crosses the vehicle's path from the farside, and the vehicle strikes the pedestrian at 50% of the vehicle's width.
<i>CarToPedestrianNearsideAdult25</i>	An adult pedestrian crosses the vehicle's path while walking from the nearside, and the vehicle strikes the pedestrian at 25% of the vehicle's width.
<i>CarToPedestrianNearsideAdult75</i>	An adult pedestrian crosses the vehicle's path while walking from the nearside, and the vehicle strikes the pedestrian at 75% of the vehicle's width.
<i>CarToPedestrianNearsideChildObstructed50</i>	A child pedestrian crosses the vehicle's path, running from behind and obstruction from the nearside, and the vehicle strikes the pedestrian at 50% of the vehicle's width.
<i>CarToPedestrianLongitudinalAdult25</i>	An adult pedestrian walking in the same direction in front of the vehicle and the vehicle strikes the pedestrian at 25% of the vehicle's width.
<i>CarToPedestrianLongitudinalAdult50</i>	An adult pedestrian walking in the same direction in front of the vehicle and the vehicle strikes the pedestrian at 50% of the vehicle's width.
<i>CarToPedestrianTurningAdult50</i>	A vehicle turns towards an adult pedestrian crossing its path, walking across a junction, and the vehicle strikes the pedestrian at 50% of the vehicle's width.
<i>CarToPedestrianReverseAdultChildMoving50</i>	A vehicle travels rearwards towards an adult or child pedestrian crossing its path walking from the nearside, and the vehicle strikes the pedestrian at 50% of the vehicle's width.
<i>CarToPedestrianReverseAdultChildStationary</i>	A vehicle travels rearwards toward an adult or child pedestrian standing still, and the vehicle strikes the pedestrian at 25, 50, or 75% of the vehicle's width.
<i>CarToBicyclistNearsideAdult50</i>	A bicyclist crosses the vehicle's path cycling from the nearside and the vehicle strikes the bicyclist.
<i>CarToBicyclistNearsideAdultObstructed50</i>	A bicyclist crosses the vehicle's path cycling from the nearside from behind an obstruction, and the vehicle strikes the bicyclist at 50% of the vehicle's width.
<i>CarToBicyclistFarsideAdult50</i>	A bicyclist crosses the vehicle's path cycling from the farside, and the vehicle strikes the bicyclist at 50% of the vehicle's width.
<i>CarToBicyclistLongitudinalAdult25</i>	A bicyclist cycling in the same direction in front of the vehicle and the vehicle strikes the cyclist at 25% of the vehicle's width.
<i>CarToBicyclistLongitudinalAdult50</i>	A bicyclist cycling in the same direction in front of the vehicle and the vehicle strikes the cyclist at 50% of the vehicle's width.
<i>CarToBicyclistTurningAdult50</i>	A vehicle turns towards a bicyclist crossing its path, cycling in the opposite direction across a junction, and the vehicle strikes the cyclist at 50% of the vehicle's width.
<i>CarToMotorcyclistRearStationary</i>	A vehicle travels toward a motorcyclist, and the vehicle strikes the motorcycle's rear.
<i>CarToMotorcyclistRearBraking</i>	A vehicle travels forward toward a motorcyclist traveling at a constant speed and then decelerates, and the vehicle strikes the rear of the motorcycle.
<i>CarToMotorcyclistFrontTurnAcrossPath</i>	A vehicle turns across the path of an oncoming motorcyclist traveling at a constant speed, and the vehicle strikes the front of the motorcycle.
<i>CarToMotorcyclistOncoming</i>	A vehicle drifts out of the lane and into the path of a motorcyclist traveling in the opposite direction in the adjacent lane.
<i>CarToMotorcyclistOvertaking</i>	A vehicle drifts out of the lane and into the path of a motorcyclist traveling in the same direction in the adjacent lane.

property for the relationship between an event and the temporal entity in which it occurs (or in which it is foreseen to occur). The main event case of our proposal is the *Collision* event class when a vehicle collides with another vehicle, pedestrian, animal, or other moving or stationary obstruction. This results in injury, disability, death, and property damage as well as financial costs to both society and the individuals involved. As collision cases can be the most varied, we have only considered the cases involving VRU here. An exhaustive list of VRU collision cases of interest is adopted according to the recommendations of The European New Car Assessment Program, which has been listed in Table 2 [7].

The listed cases incorporate the main groups of collision scenarios based on real accident databases [41]. Vehicle ma-

neuvers and conflict situations are identified in the accident database to create these groups of collisions. These scenarios are used by protocols to test collision avoidance mechanisms in new driver assistance systems. Two major scenarios are generally addressed: crossing accidents between the VRU and the same nearside or farside vehicles and longitudinal accidents where the VRU and the vehicles move in the same direction. Therefore, we understand that the same scenarios are the ones that our DT-RI must anticipate to act on the road intersection as a whole and avoid or mitigate the effects of a collision between vehicles and VRUs. Furthermore, these scenarios and corresponding risks to VRUs can even assist [integrated ethical decision-making mechanisms](#), e.g., [42], when choosing a sequence of actions for minimizing harm

in unavoidable collisions [43].

Furthermore, note in Fig. 5 that the weather conditions are closely linked in our dynamical model with the collision scenarios discussed above. Environment-aware digital twins in critical domain applications have recently been highlighted [44]. Weather extremes become more frequent and severe, and the importance of building weather, climate, and environmental information elements into digital twins of critical systems such as cities, ports, flood barriers, energy grids, and transport networks increases. In this way, the *Weather* class has been defined for local weather conditions in the neighboring area of the road intersection. The weather describes the state of the atmosphere at a particular place and time, collecting features of interest such as precipitation, temperature, humidity, visibility, and wind, all of them of extreme importance in modeling vehicle movement characteristics, given that they directly influence the general perception of the road intersection for traffic participants and how the participants themselves (or cyber-physical systems) act on the road intersection.

D. INTERSECTION TRAFFIC MODELING

The traffic model adopted in the DT-RI framework is of a microscopic type, which enables the simulation of the individual behavior of each vehicle and VRU, as well as their interactions. This approach considers key factors such as speed, acceleration, and spatial positioning within the road intersection under analysis. The traffic model is supported by both static and dynamic data inputs, as previously described. Furthermore, the intersection traffic model incorporates all permitted movement groups for both vehicles and VRUs, based on the intersection's geometric configuration. This information is defined and validated by traffic engineering specialists.

Figure 6(a) presents an example of a four-way road intersection with two lanes per approach. It illustrates the permitted movement groups for vehicles on road r_1 , which is taken as the source point for the declaration of movements. Since there are two lanes, we define two main movement groups $G_1^{r_1}$ and $G_2^{r_1}$, which group together movements that define the relationship between source and destination lanes at the intersection, where $(m_{1,1}^{r_1}, m_{1,2}^{r_1}, m_{1,3}^{r_1}) \in G_1^{r_1}$ and $(m_{2,1}^{r_1}, m_{2,2}^{r_1}, m_{2,3}^{r_1}) \in G_2^{r_1}$. Similarly, the traffic model defines the permitted movements for VRUs (Vulnerable Road Users), based on the road crossing trajectories for VRUs computed in [45]. Continuing with our intersection example, Fig. 6(b) shows the most likely permitted movements for two VRUs: vru_1 is a pedestrian, and vru_2 is a cyclist, with $(m_{1,1}^p, m_{1,2}^p) \in G_1^p$ and $(m_{1,1}^c, m_{1,2}^c) \in G_1^c$. Pedestrians (VRUs of pedestrian type) walk along the sides of each road and typically cross near each corner of the intersection. Cyclists, on the other hand, ride along the sides of the roads using dedicated bike

lanes where available, or in areas close to the curbs when such lanes are not present.

The so-called What-if Analysis framework, which is made possible by DTs, in our case, must first determine the type of collision scenario based on the agents present, their distribution in the geometry of the road intersection, and their movement dynamics given the prevailing weather conditions. Once the group of potential collision scenarios has been determined, the DT-RI must make use of the other features of interest, such as the speed of the agents, to foresee, through simulations, any future scenario that may result in a collision with the system returning, whenever possible, the collision point and the time to collision (TTC) [7].

With the road intersection traffic model, supported by the semantics described for the construction of the road intersection digital twins, the objective functions for the system were defined. According to the movement groups defined in the traffic model, vehicles and VRUs may have various conflict relationships that define conflict zones within the road intersection, resulting from the movements of all system agents, as shown in Fig. 6(c). These conflicts can result in collisions at a specific time t_c . The common types of collisions for road intersection scenarios are classified as Farside collision, Nearside collision, Nearside obstructed collision, Longitudinal collision, farside turning collision and nearside turning collision, all of which are listed in detail in the collision class Table 2, and graphically represented in Fig. 6(d), 6(d), 6(d), 6(d), 6(d), and 6(d) respectively.

Given that our objective is to improve the safety of vulnerable agents near road intersections, our objective function is to minimize vehicle and vulnerable agent collisions as modeled in Equation 1. Given all possible conflict relationships $R(v, vru)$ identified within the road intersection, the objective function in the DT-RI system is to minimize the collision risk C_i and increase the collision time TTC_i so that actions can be taken proactively to minimize the impact or eliminate potential collisions.

$$\min \sum_{i=1}^{R_i(v, vru)} \left(\frac{1}{TTC_i} + C_i \right) \quad (1)$$

Considering that each agent has a movement model defined by the trajectory $p(t)$ from Eq. 2, represented in the collision cases of Fig. 6 by $p_v(t)$ y $p_{vru}(t)$ for vehicles and VRUs, respectively, then a collision occurs, as per Eq. 3, when the evaluated trajectories for the collision time are less than the radius d_c of a conflict area represented by a red dashed circle in Fig. 6. Thus, the collision time can be computed according to Eq. 4.

$$p(t) = p_o + v_o t + a_o t^2 \quad (2)$$

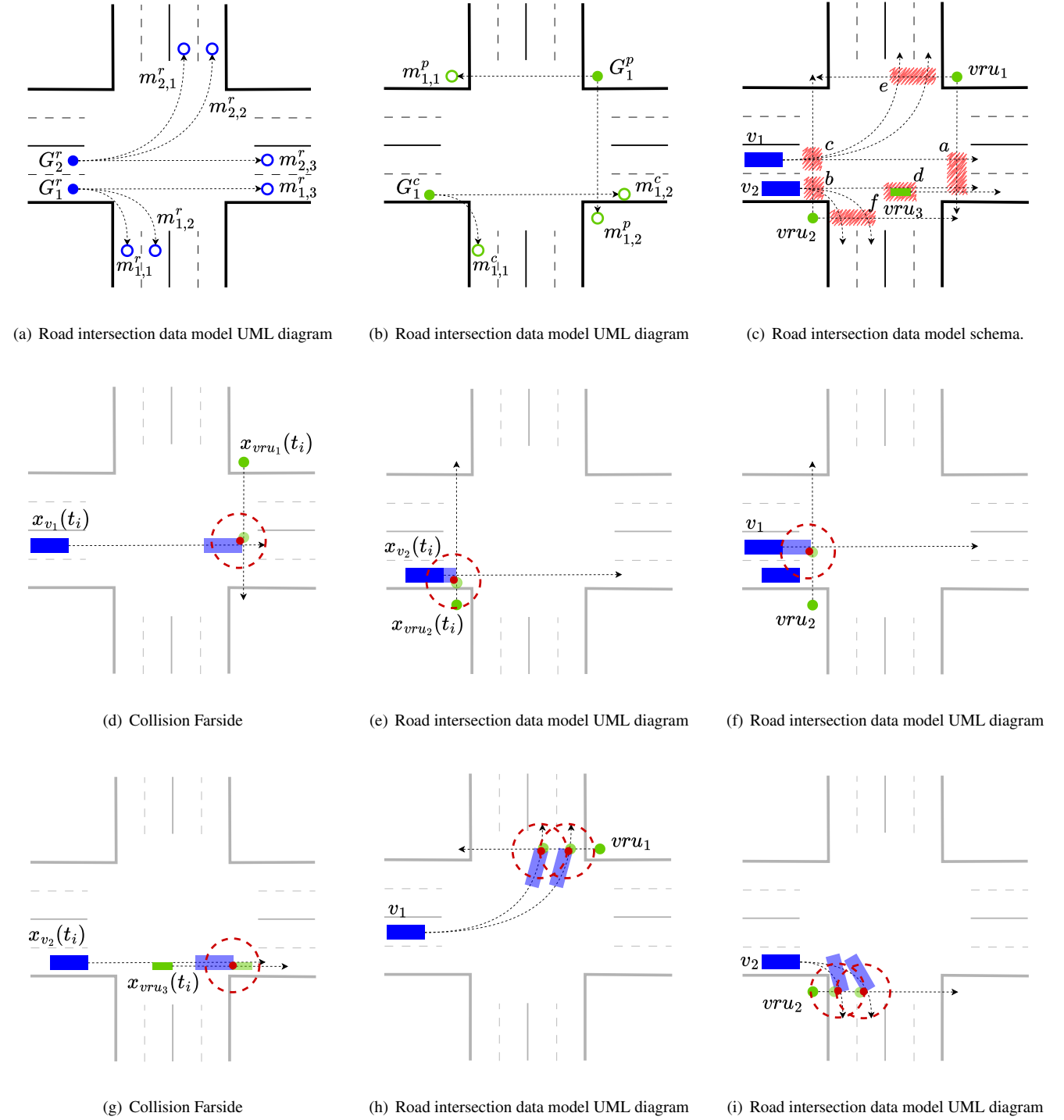


FIGURE 6. Road Intersection Traffic Model

$$C_i = \begin{cases} \|p_v(t_c) - p_{vru}(t_c)\| \leq d_c, & t \in [t_0, T_{max}] \\ 0, & \text{others} \end{cases} \quad (3)$$

$$TTC_i = \int_{t_0}^{T_{max}} \|p_v(t) - p_{vru}(t)\| dt \quad (4)$$

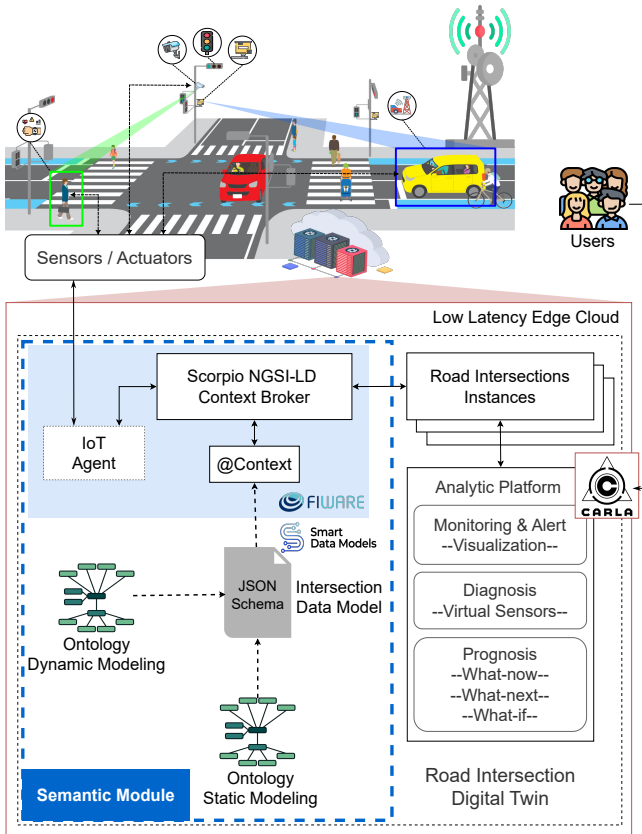


FIGURE 7. Semantic Module Implementation.

IV. A FULLY OPEN-SOURCE-BASED INSTANTIATION

The ontology's purpose is solely to give semantics to the data collected at the road intersection. Thus, this section needs a further step toward building a corresponding virtual representation (i.e., computational models) of the physical assets and processes at road intersections. Moreover, such a computational model also requires well-defined methods for synchronization and interaction tasks with physical world assets through a communication channel, as illustrated in Fig. 1. Creating data models from scratch can be an inefficient approach for prototyping, considering that there are already solutions to provide support for creating and operating DTs, which are usually known as DT platforms. Lehner et al. [46] do a comparative analysis on Amazon, Eclipse ecosystem, and Microsoft Azure DT as platforms to support the development of prototypes.

Our objective here is to demonstrate a methodology for creating DT-RI prototypes using an ecosystem built on open-source tools and data repositories, enabling inexpensive and accessible prototyping of ontology-based semantic modules, as highlighted in Section III. This process involves platforms for creating and instantiating smart data models, as illustrated in Fig. 3. Furthermore, these computational instances must be populated with physical-world data. Finally, a tool for DT instantiation is required to allow users to interact effectively

with the physical world.

A. SMART DATA MODEL AND APPLICATION PROGRAMMER INTERFACE

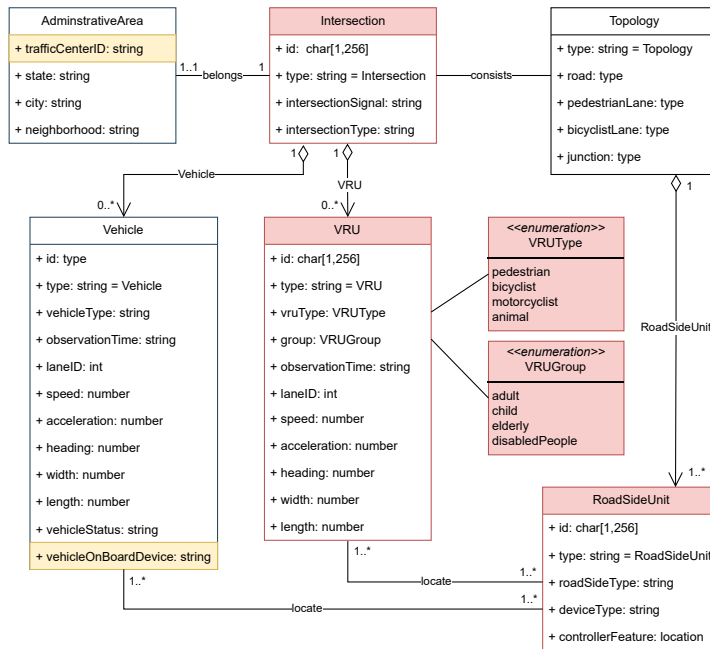
For implementing the DT-RI semantic module, we have chosen the FIWARE platform, as shown in Fig. 7, which provides all the necessary tools for building DTs. FIWARE is containerized, highly scalable, and easy to implement and replicate [14]. Cloud-based network-slicing solutions [47] can be designed to meet the computational and communication network requirements of the elements shown in Fig. 7. Data originates from heterogeneous sources, utilizing diverse protocols and encompassing varying data attributes, types, and relationships. Consequently, it is essential not only to standardize communication among DT components but also to ensure that the data formats flowing through these components are standardized to guarantee interoperability. IoT Agents in Fig. 7, translate IoT-specific protocols from the sensing and acting system of the road intersection into the NGSI-LD context information protocol.

FIWARE provides data modeling and management mechanisms through the FIWARE RESTful application programmer interface (API), which allows access to current information on physical systems and data updates. The main element in the NGSI-LD standard is the context entities that represent the physical or logical objects. Each context entity has a set of attributes, and attribute metadata represents real-life properties. Based on context entities, NGSI-LD provides a smart data model for context information, an interface for exchanging data, and an interface for exchanging information on how to obtain such data. NGSI-LD API interactions are made using HTTP/HTTPS requests. The Scorpio context broker in Fig. 7, is the core element that allows managing and requesting context information in NGSI-LD format.

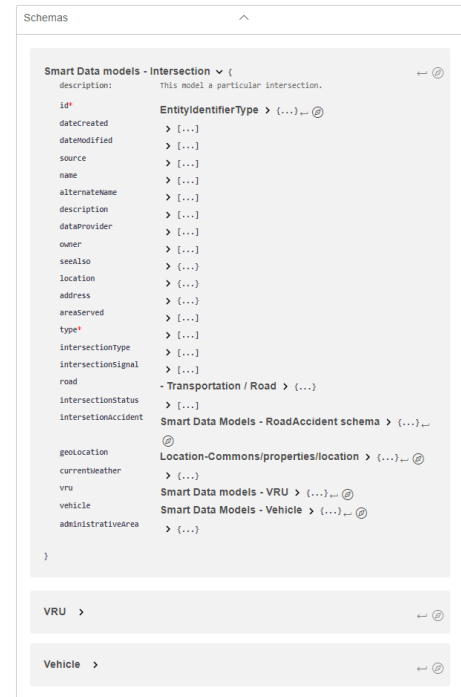
We have followed the recommendations in [13] to create our smart data model for a road intersection ontology proposed in Section III. We have created a smart data model called *Intersection* as shown in the road intersection data model UML diagram in Fig. 8(a). Relationships between entities make up the road intersection. Based on the intersection smart data model, the schemas for the entities are created, where the technical representation of the model defines the technical data types and structure on a written document for human readers. Figure 8(b) shows the main schema of the data model of the crossing together with the schemas of the other main entities involved in the construction of the digital twin.

Our smart data model reuses different models from the FIWARE Smart Data Models catalog³. The Smart Data Models provide data models for digital twins and data spaces, enabling true data interoperability between diverse systems based on open-licensed data models. They are compatible with the FIWARE platform but also export in several other

³<https://smartdatamodels.org/>



(a) Road intersection data model UML diagram



(b) Road intersection data model schema.

FIGURE 8. Road Intersection Data Model

formats, such as JSON, JSON-LD, CSV, SQL, and DTDL. Specifically, we have reused the data models belonging to the transportation domain to create the smart intersection data model. Other critical elements for developing VRU safety applications were modeled from scratch. The *VRU* and *RoadSideUnit* models were created specifically for use in our road intersection model. The VRU models specify features of interest for safety applications such as age, location, travel speed, or direction, which the road intersection sensing mechanisms could provide. In addition, the specific attributes of the intersection were created, such as *intersectionType*, *intersectionStatus*, *intersectionSignal*, among others related to the transportation system. The new models we created have been highlighted in red, while new properties of existing models have been highlighted in yellow in Fig. 8(a).

Furthermore, a context is necessary when using these smart data models under a linked data paradigm. Therefore, every implementation needs a *context.jsonld* reference file (seen interacting with JSON Schema Fig. 7), used by the context broker for instantiating the road intersection digital twin. The FIWARE community offers the necessary mechanisms using the Swagger tool⁴ to generate the NGSI-LD context file from the road intersection smart data model.

B. POPULATING THE SMART DATA MODEL

Custom agents, created by the sensing system developers for FIWARE IoT, can also be easily integrated to interact with the context broker. The communication mechanisms and protocols used in the data acquisition phase are outside the scope of this work, so we assume that the IoT agents in Fig. 7 provide reliable and secure data for creating the DT-RI instances.

Third parties responsible for the operation and management of the road intersections must develop their IoT agents⁵ or reuse the IoT agents already available⁶. For prototyping purposes, we have populated our smart data model with the inD dataset, a collection of traffic recordings from four distinct locations captured via drone, detailing each road user's movement and type [48]. Given the detailed level of user information that the dataset provides, for this proof of concept, we will assume that the dataset is equivalent to the information that IoT agents should provide from sensing the road intersection scenario.

We have chosen the ASAM OpenDRIVE standard [49] for implementation, an open-source format organized in a hierarchical structure and compiled in Extensible Markup Language (XML). OpenDRIVE allows the modeling of the static information of the road intersection and provides a standardized format for road descriptions, reducing the cost

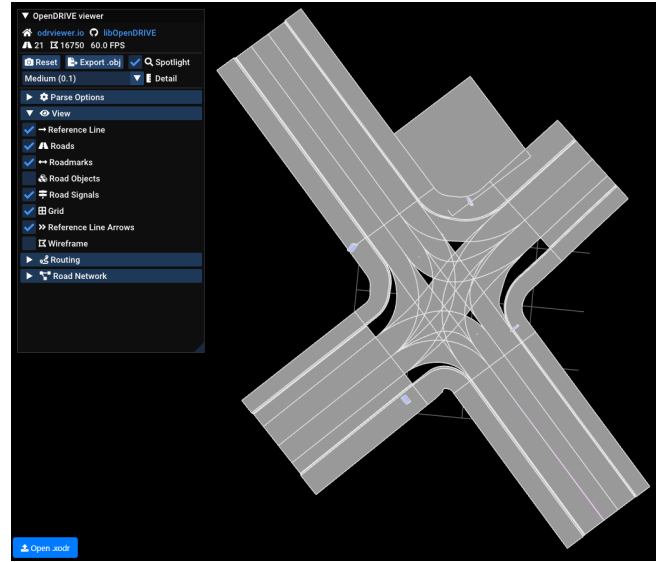
⁴<https://swagger.io/>

⁵<https://iotagent-node-lib.readthedocs.io>

⁶<https://www.FIWARE.org/catalogue>



(a) Road intersection view



(b) Opendrive representation

FIGURE 9. Road Intersection Static Information Modeling

of creating and converting these files for their development and testing purposes. The implementation is very complex, as roads are generated from a sequence of segments.

To construct the OpenDRIVE file, we use tools designed for creating and editing OpenDRIVE structures, some of which can be derived from widely recognized map formats. To simplify the development of virtual maps using OpenDRIVE, the HLRS is developing the software ODDLOT⁷. The ODDLOT tool allows one to load maps and draw roads in a graphical interface using different prototypes specified by the number of lanes, width, and road markings. The ODDLOT tool provides a graphical interface for loading maps and designing roads using different prototypes specified by the number of lanes, width, and road markings. Junctions can be created automatically by defining the intersection area or manually using a specialized editor.

For a straightforward approach toward the instantiation of a road intersection virtual representation, we also can take advantage of the fact that the inD dataset provides us both OpenDRIVE files and images of the road intersection, as can be seen in Figures Fig. 9(b) and 9(a), respectively [48]. The static information of our smart data model for the road intersection can be properly visualized thanks to the OpenDRIVE Online Viewer tool⁸.

The context broker in Fig. 7 provides the most up-to-date context information for the described entities. Other components subscribe to updates for the entities they are interested in. Dynamic information, such as data related to vehicles, vulnerable road users (VRUs), and weather conditions, is

processed by the context broker. This data is obtained through IoT agents, ensuring seamless integration and real-time updates. To represent the scenario's dynamic elements, we have selected recording number 7 from the inD database, and within it, we have chosen the initial frame. At that moment, 8 traffic agents were present at the road intersection, 6 vehicles (5 parked), and 2 cyclists [48].

The instantiated road intersection entity is shown in Listing 1 using normalized NGSI-LD format. The relationships with the dynamic elements, the vehicle entities, and the VRU entities are declared within the road intersection entity. Due to space constraints, we have only represented the entity of the first VRU and the first vehicle in the selected frame, in Listing 2 and Listing 3 respectively, also using normalized NGSi-LD format. The properties specified with values in each entity result from the data provided in the InD dataset, which would be similar to the data provided by the sensors in the roadside units at the intersection illustrated in Fig. 2.

```
{
  "id": "urn:ngsi-ld:Intersection:001",
  "type": "Intersection",
  "intersectionSignal": {
    "type": "Property",
    "value": "non-signalized"
  },
  "intersectionType": {
    "type": "Property",
    "value": "four-leg"
  },
  "geoLocation": {
    "type": "GeoProperty",
    "value": {
      "type": "Point",
```

⁷<https://www.hlrs.de>

⁸<https://odrviewer.io/>

```

        "coordinates": [50.78207, 6.0711
            ],
    },
    "vru": {
        "type": "Relationship",
        "object": "urn:ngsi-ld:VRU:001"
    },
    .
    .
    .
    "vehicle": {
        "type": "Relationship",
        "object": "urn:ngsi-ld:Vehicle:001"
    },
    .
    .
    .
    "@context": [
        "https://mygithub/
        FIWARE_smart_intersection/
        datamodels.context-ngsi.jsonld"
    ]
}

```

LISTING 1. Road Intersection Entity

```

{
    "id": "urn:ngsi-ld:VRU:001",
    "type": "VRU",
    "vruType": {
        "type": "Property",
        "value": "bicyclist"
    },
    "heading": {
        "type": "Property",
        "value": 130.52964
    },
    "Location": {
        "type": "Relationship",
        "object": "urn:ngsi-ld:RoadSideUnit:
            001"
    },
    },
    "speed": {
        "type": "Property",
        "value": 4.9064
    },
    "acceleration": {
        "type": "Property",
        "value": 0.8421
    },
    "@context": [
        "https://mygithub/
        FIWARE_smart_intersection/
        datamodels.context-ngsi.jsonld"
    ]
}

```

LISTING 2. VRU Entity

```

{
    "id": "urn:ngsi-ld:Vehicle:001",
    "type": "Vehicle",

```

```

    "vehicleType": {
        "type": "Property",
        "value": "car"
    },
    "heading": {
        "type": "Property",
        "value": 130.43668
    },
    "width": {
        "type": "Property",
        "value": 1.93507
    },
    "length": {
        "type": "Property",
        "value": 4.94025
    },
    "Location": {
        "type": "Relationship",
        "object": "urn:ngsi-ld:RoadSideUnit:
            001"
    },
    "speed": {
        "type": "Property",
        "value": 9.6565
    },
    "acceleration": {
        "type": "Property",
        "value": 2.0056
    },
    "vehicleStatus": {
        "type": "Property",
        "value": "moving"
    },
    "@context": [
        "https://mygithub/
        FIWARE_smart_intersection/
        datamodels.context-ngsi.jsonld"
    ]
}

```

LISTING 3. Vehicle Entity

C. VIRTUAL ROAD INTERSECTION VISUALIZATION: SCENARIOS AND SIMULATIONS

As depicted in Fig. 1, DTs have a Virtual Representation with computational models to provide a unified platform for simulation, learning, prediction, and visualization. This Analytic Platform is detailed in Fig. 7, interacting with the instantiated FIWARE DT's Semantic Module (via the Scorpio NGSI-LD Context Broker). Such interaction with the smart data model discussed above is needed not only for performing basic Monitoring and Alert Visualization as in regular Traffic Control Centers but also for advancing toward areas such as simulations for Diagnosis (e.g., use of Virtual Sensors) and Prognosis studies (i.e., What-now, What-next, What-if scenarios) in future ITS using DT-RI.

Note that realistic and flexible visualization is a basic feature required for dealing with current state observations and

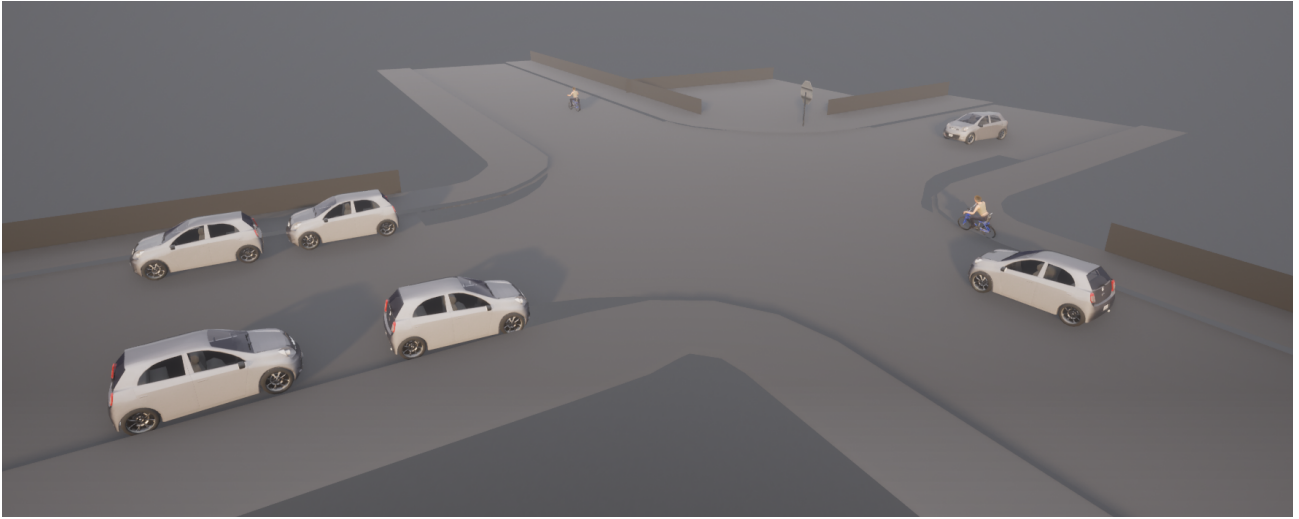


FIGURE 10. DT-RI Visualization and Simulation using CARLA: 6 vehicles, and 2 cyclists (frame 0 of recording 7 from inD database).

possible future scenarios in simulated cases. CARLA⁹ is an open-source driving simulator grounded on Unreal Engine¹⁰, that implements a scalable client-server architecture.

In its simplest simulation mode, the CARLA platform allows the creation of the layout of the road intersection using the OpenDRIVE specification and representing all the agents in the traffic scenario. For this, CARLA clients were developed to serve as an interface between the Scorpio context broker (seen in Fig. 7) and the CARLA server, where simulations and their visualization are executed.

CARLA clients collect information corresponding to each entity instantiated by the broker using the NGSI-LD API and insert the information into the simulator through the CARLA Python API to represent vehicles and VRUs on the previously generated map. Unlike previous uses that integrate the CARLA simulator as a virtualization element in creating the DT, e.g., [50], our proposal focuses on a FIWARE-centric approach. In our case, the CARLA simulator illustrates just one of the many applications that could consume the context data from the Scorpio broker. Thus, other DT's enablers native to FIWARE can be implemented to store historical data for training machine learning algorithms as well as interacting with the simulations executed in CARLA.

The reference intersection of the inD database that we have used to instantiate the FIWARE entities of the DT-RI has also been created as a world in the CARLA simulator from the OPENDRIVE file available in the database. Building on the methodology used to instantiate FIWARE entities, a simulation was conducted in CARLA, replicating recording 7 of the inD database. **At that time, the road intersection featured eight traffic agents, including 6 moving vehicles and 2 cyclists [48]. This scenario, along with the agents, was retrieved by CARLA through Scorpio, as defined in Listing 2 and Listing**

3. At that time, recording 7 depicts an urban intersection with eight traffic agents: two moving vehicles, four parked vehicles, and two cyclists [48]. Despite the low number of moving agents, the scenario is relevant for studying VRU-vehicle interactions in a controlled and computationally efficient environment. CARLA's clear visualization supported visual validation and analysis, laying a solid foundation for future, more complex scenarios. This scenario, along with the agents, was retrieved by CARLA through Scorpio, as defined in Listing 2 and Listing 3.

Figure 10 shows a textured visualization of that scene using CARLA. Thus, the simulation at the beginning, corresponding to frame 0 of recording 7, the same that was used to create the FIWARE entities in subsection IV-B. This simulation can even be visualized remotely (but without texture) in a Web browser using the CarlaViz¹¹ tool to provide simulation as a service to remote DT-RI users, as shown in Fig. 7.

V. CONCLUSIONS

With the advancement of information technologies in transportation, Digital Twin (DT)-enabled applications are poised to transform the management and operations of Intelligent Transportation Systems (ITS). However, their primary focus often remains on efficiency, rather than on the crucial safety of Vulnerable Road Users (VRUs). Addressing this gap requires a robust information modeling framework that effectively integrates the physical world with cyberspace in complex and dynamic environments such as road intersections.

This work proposed, for the first time, an ontology-based information modeling method for road intersections using open-source and market-driven platforms to enable DT-based VRU safety applications. The ontology plays a central role in building semantic-level models that accurately represent the entire intersection environment within a virtual space.

⁹<https://carla.org/>

¹⁰<https://www.unrealengine.com/>

¹¹https://carla.readthedocs.io/en/latest/plugins_carlaviz/

It comprehensively defines the properties required to create a Road Intersection Digital Twin (DT-RI), addressing the semantic complexity of these scenarios through a structured representation of their components and interactions. This includes both static elements (geometric infrastructure, road components, sensors, and actuators) and dynamic elements (traffic agents like vehicles and VRUs, their behavior, and environmental conditions such as weather), as well as their relationships.

Within this framework, the structured information supports accident reduction through high-fidelity monitoring. The DT-RI functions as a continuously evolving, sensor-fed virtual representation that provides an accurate understanding of the current state of the intersection ("what-now"). Real-time data obtained from deployed sensors feeds the DT-RI, which, in turn, leverages its semantically enriched database in the Model layer to execute computational models—including AI/ML techniques. These models, trained with historical and real-time data, allow the DT to analyze and predict traffic behavior.

Simulation is a key component of the Virtual Representation layer. Unlike traditional simulations that often diverge from reality due to outdated parameters, the DT-RI integrates real-time data for more accurate and dynamic simulations. This enables "what-if" scenario analysis and supports optimization techniques to enhance safety. Predictive analytics further empower the DT-RI to anticipate potential conflicts and collisions, particularly involving VRUs, based on agents' positions, speeds, behaviors, and environmental conditions. The ontology defines specific collision event classes, including VRU cases recommended by Euro NCAP, which the DT-RI is designed to foresee.

Additionally, the DT-RI calculates critical safety metrics such as Time-to-Collision to assess the imminence and severity of potential accidents. Optimization algorithms can then determine the best strategies to reduce risks. Real-time simulations and predictions improve over time, enhancing both accuracy and decision-making effectiveness. Based on these predictions and risk calculations, the system can proactively activate actuators via V2X communication to prevent or mitigate collisions. The DT-RI's objective is to minimize collision risk and maximize time-to-collision to allow for proactive interventions.

Consequently, the detailed predictive capabilities of the DT-RI enable traffic authorities to implement more precise traffic control measures, including adaptive and dynamic smart traffic light systems. These systems leverage DT predictions related to traffic density, agent behavior, and potential conflicts to optimize both safety and efficiency at intersections.

Importantly, the ontology-driven approach contributes to road safety beyond individual intersection modeling. By embedding semantic understanding within the DT-RI, the system

gains contextual awareness of the environment, allowing it to reason over data and adapt behavior based on specific conditions—such as identifying school zones or accident-prone areas under adverse weather. This semantic layer also fosters interoperability among heterogeneous ITS components through standardized communication protocols, enhancing the coordinated response to safety-critical events.

Furthermore, the ontology-driven nature of the DT enables sophisticated simulation and risk analysis, where causal relationships among agents and infrastructure are clearly defined and machine-interpretable. These capabilities directly support real-time decision-making, enhance the situational awareness of autonomous vehicles, and facilitate safer automated navigation. The system's ability to continuously learn and evolve based on new data ensures ongoing refinement of safety strategies, making it resilient and adaptable to emerging traffic patterns and behaviors.

Future work will focus on two fundamental directions to enhance the DT-RI: improving both computational and communication-network capabilities of the system components. To address scalability across various ITS elements, infrastructure must align with FIWARE's highly scalable features. Therefore, a suitable cloud-based network-slicing solution must be explored, particularly for ensuring security and privacy throughout the system's operational lifecycle. Furthermore, implementation of a data-format standardization block—depicted in Fig. 7—will be pursued. This block will enable translation of IoT-specific protocols from the sensing and actuation systems at the intersection into the NGSI-LD context information protocol.

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